Analytical Chemistry Cumulative Examination
November 1, 2024

Topic: Basic Quantitative Analysis Using GC-MS: Terminology, Theory and Use of Equations for Separations, and Electron Ionization Mass Spectrometry

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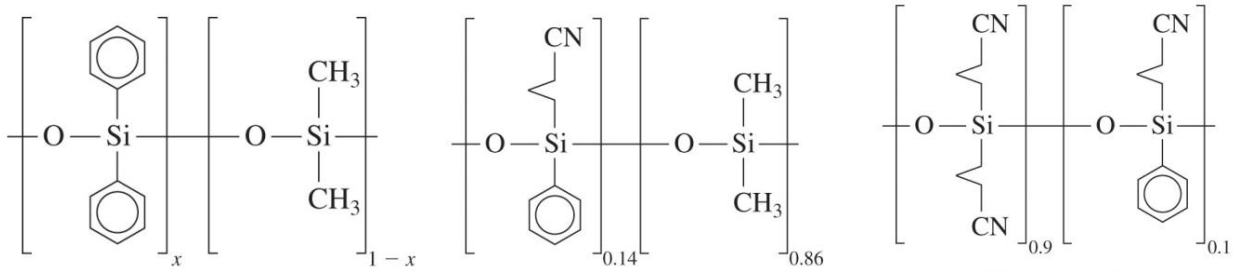
1. Some terminology for GC and MS

A. Assume that a small organic molecule is analyzed by electron ionization mass spetrometry. Consider the resulting mass spectrum.
(a) Give the definitions for "base peak" and "molecular ion". (b) When is the base peak the same as the peak for the molecular ion?

B. In chromatography, what is the plate height? Also, how does it relate the length of the column to the peak width?

2. Stationary phase

Below are chemical structures (A, B, and C) of some common stationary phases that are used in GC.
(a) Place these in order from most polar to least polar.
(b)Which of these would be the most logical choice for separating analytes that are very nonpolar.



(Diphenyl)$_x$(dimethyl)$_{1-x}$ polysiloxane

A

(Cyanopropylphenyl)$_{0.14}$ (dimethyl)$_{0.86}$ polysiloxane

B

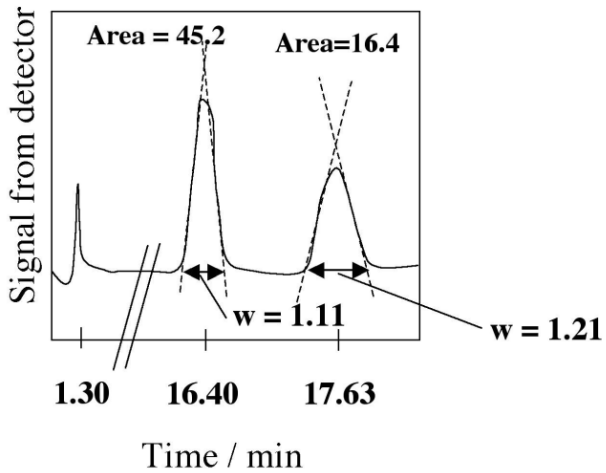
(Biscyanopropyl)$_{0.9}$ (cyanopropylphenyl)$_{0.1}$ polysiloxane

C

3. Efficiency.

Below is a list of parameters that can be changed to vary the efficiency of the chromatographic separation of a mixture of solutes. Explain how you would modify each (increase, decrease, both, or neither) to optimize a chromatographic separation of a mixture of solutes. Explain why that change is expected to improve the efficiency.
(a) Diameter of the capillary column
(b) Temperature of the column
(c) Thickness of the stationary phase on the walls of a capillary column
(d) Length of the column
(e) Flow velocity of the mobile phase

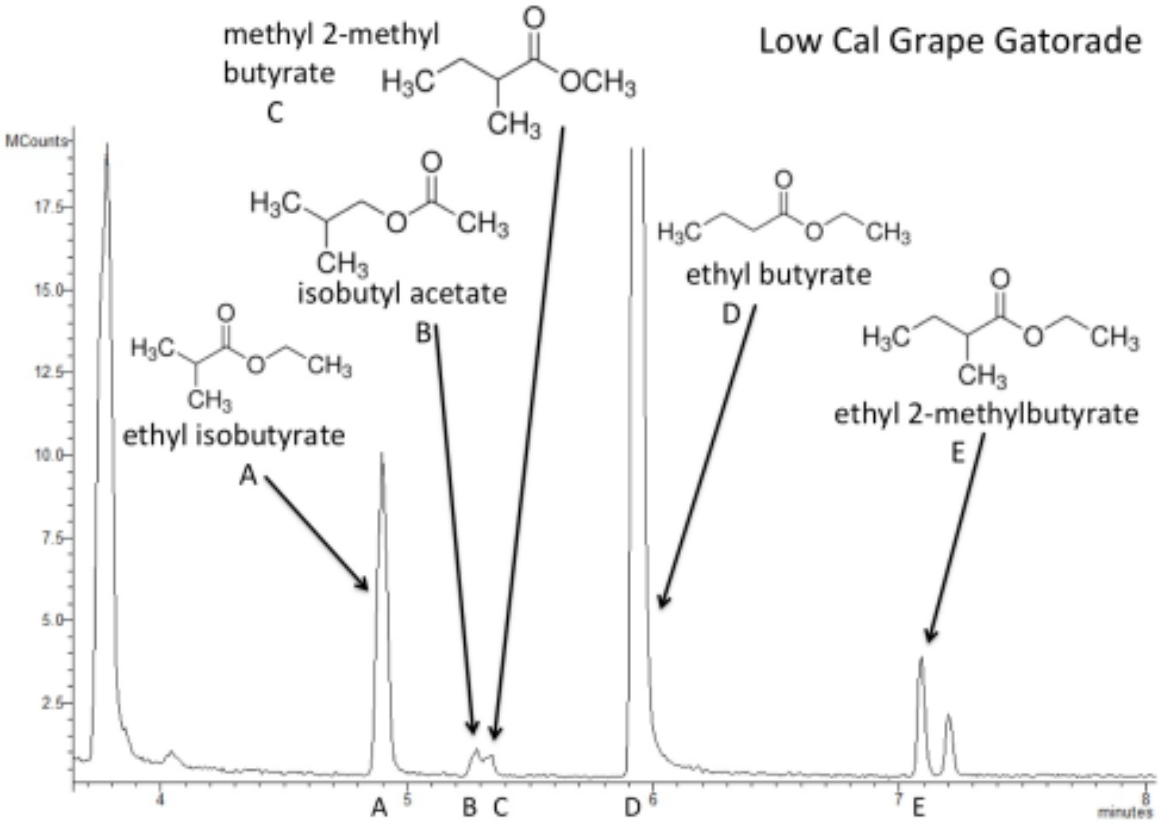
4. Quantitation of solutes from chromatogram. Below is a chromatogram obtained from the separation of a mixture of A (at 16.40 min) and B (at 17.63 min). Areas under the peaks are given in the figure.



Calculate the relative percentages of A and B in the mixture. (You will need the detector response factor to calculate the answer. Thus, in a separate experiment it was found that a solution containing an equimolar concentration of A and B, produced a detector response for A of 0.385 and one for B of 0.615.)

5. GC-MS data and calculation of important parameters. Below is a portion of a GCMS chromatogram of the headspace taken over Low Cal Grape Gatorade sports drink. In the chromatogram, it is the total ion signal from the mass spectrometry detector that is plotted as a function of time. The length of the column is **30.0 m** and flow rate is **1.00 L/min**. Information about the mobile phase and six of the peaks is summarized in the table below:

Chemical Species	Simple label to refer to these compounds	Reten -tion time, min	Peak width, min	Molecular mass, g/mol
Mobile phase, helium	--	1.47	--	--
Ethyl isobutyrate	A	4.90	0.10	116.16
Isobutyl acetate	B	5.27	0.10	116.16
Methyl 2-methyl butyrate	C	5.33	0.10	116.16
Ethyl butyrate	D	5.95	0.10	116.16
Ethyl 2-methylbutyrate	E	7.10	0.10	130.18



(a) Which solute, A, C, or E, has the largest partition coefficient (distribution constant)? Why?

(b) Calculate the number of plates for the column based on the retention time and peak width for (A) ethyl isobutyrate.

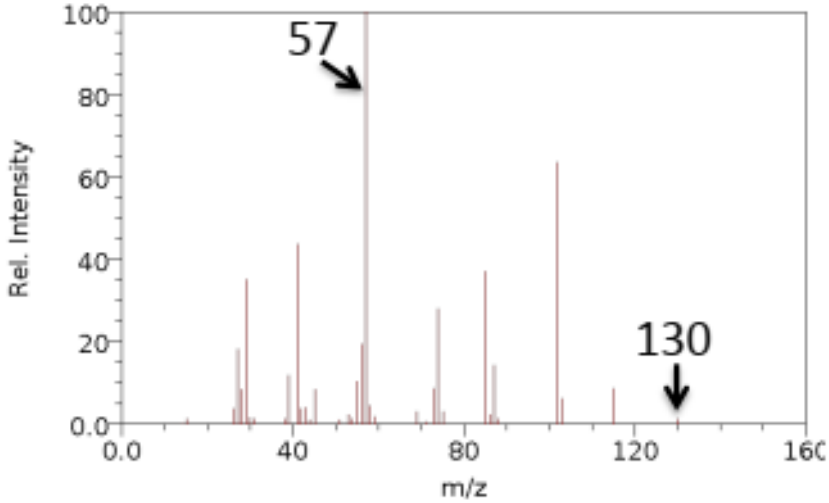
(c) What is the plate height for the column, given the number of plates determined in part (b)?

(d) Calculate the resolution for (B) Isobutyl acetate and (C) Methyl 2-methyl butyrate.

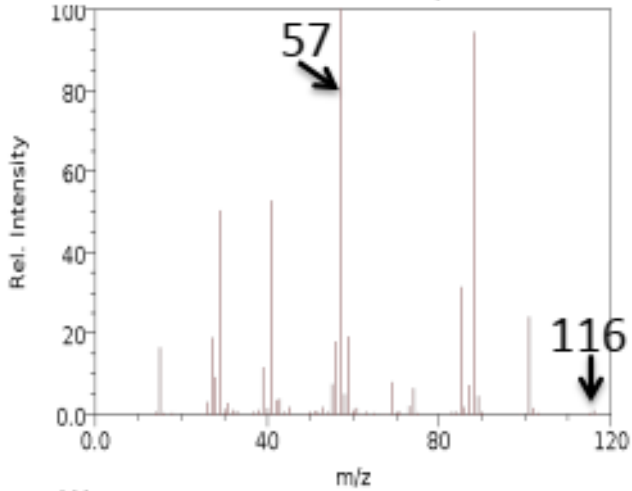
(e) What is the **new retention time for (C) Methyl 2-methyl butyrate** when the resolution for (B) isobutyl acetate and (C) methyl 2-methyl butyrate is improved to **1.5**? (To perform the calculation, you will need the original resolution that you calculated in (d) above and the original retention time for (C) found in the table.)

(f) Match each of the species, (**C**) methyl 2-methyl butyrate, (**D**) ethyl butyrate, and (**E**) ethyl 2-methylbutyrate to its electron ionization-mass spectrum. ***Refer to table within this problem for molecular mass.*** (Write the letter for the compound directly on its spectrum.) *The presence of peaks at m/z 43, 57, 60, 71, 116, and 130 have been labeled for you. These are key peaks that can be used to distinguish one compound from the others.*

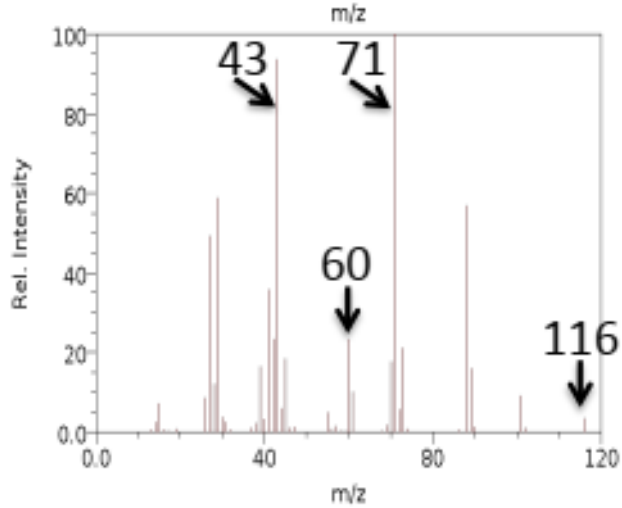
(1)



(2)



(3)



NIST Chemistry WebBook (http://webbook.nist.gov/chemistry)

Constants, Equations, and Tables for the Analytical Chemistry Cumulative Exam

$$K_A = \frac{C_{s,A}}{C_{m,A}}$$

$$k'_A = \frac{moles_s}{moles_m} = \frac{C_s V_s}{C_m V_m} = K_A \frac{V_s}{V_m} = \frac{(t_{r,A} - t_m)}{t_m}$$

$$\alpha = \frac{t_{r,B} - t_m}{t_{r,A} - t_m} = \frac{t'_{r,B}}{t'_{r,A}} = \frac{k'_B}{k'_A} = \frac{K_B}{K_A} \quad \text{where } t_{r,B} > t'_{r,A}$$

$$R_s = \frac{2(t_{r,B} - t_{r,A})}{w_A + w_B} = \frac{\Delta t_r}{w_{av}}$$

$$\sigma^2 = H\,x$$

$$\sigma^2 = H\,L$$

$$N = \frac{L}{H}$$

$$N = \frac{L}{H} = \frac{L^2}{\sigma^2} = \frac{16\,L^2}{w^2}$$

$$N = \frac{16\,t_r^2}{w^2} = \frac{5.55\,t_r^2}{w_{1/2}^2}$$

$$R_s = \frac{\sqrt{N}}{4}\left(\frac{\alpha - 1}{\alpha}\right)\left(\frac{k'_b}{1 + k'_{av}}\right)$$

$$R_s = \frac{\sqrt{N}}{4}(\gamma - 1) \text{ where } \gamma = \frac{t_{r,B}}{t_{r,A}} \text{ and where } t_{r,B} > t_{r,A}$$

$$t_{r,B} = \frac{16\,R_s^2\,H}{u}\left(\frac{\alpha}{\alpha - 1}\right)^2\left(\frac{(1 + k'_b)^3}{(k'_b)^2}\right)$$

$$H \cong A + \frac{B}{u} + C\,u$$

$$H \cong \frac{B}{u} + C_s\,u + C_m\,u$$